MTH211 : Theory of Statistics.

Lecture 28: Testing in one-parameter exponential family

Recall The one-parameter exponential family $\mathcal{P} = \{f_\theta : \theta \in \Theta\}$ with $f_\theta(x) = h(x)\, c(\theta) \exp(Q(\theta) T(x))$, $h(x) > 0$, $c(\theta) > 0$, $Q(\theta) \uparrow$ (strictly) in θ has MLR property in $T(x)$.

Then MP(α) test for $H_0 : \theta = \theta_0$ ag. $H_A : \theta = \theta_1$ ($> \theta_0$) is

$$\phi(T(x)) = \begin{cases} 1 & \text{if } T(x) > c \\ \gamma & \quad\quad = \\ 0 & \quad\quad < \end{cases}$$

(Follows directly from the NP lemma, where the corr. critical region is $\frac{f_{\theta_1}(x)}{f_{\theta_0}(x)} \geq k$, using ~~fo~~ the particular form of $f_\theta(x)$ & the fact that it has MLR in $T(x)$, this becomes equivalent to $T(x) \geq c$ etc.)

c & γ are obtained from size-α condition,
$$P_{\theta_0}(T(x) > c) + \gamma P_{\theta_0}(T = c) = \alpha$$
(modify appropriately for contns distn case)

The test ϕ is indep. of the choice of θ_1 ($> \theta_0$), & hence is UMP(α) for $H_0 : \theta = \theta_0$ ag. $H_A : \theta > \theta_0$.

Note, ϕ is a non-decreasing fn of T, further $Q(\theta)$ is increasing in θ

$\Rightarrow$ the power fn $\beta_\phi(\theta) = E_\theta \phi(T)$ is non-decreasing in θ.

Hence ϕ is UMP(α) test for $H_0 : \theta \leq \theta_0$ ag. $H_A : \theta > \theta_0$.

What if $Q(\theta)$ (in $f_\theta(x)$) $\downarrow$ (strictly) in $T(x)$? Then MLR property in $-T(x)$ and for the same set of hypotheses, ie, $H_0 : \theta \leq \theta_0$ ag. $H_A : \theta > \theta_0$, the UMP(α) test is given by

$$\phi(T(x)) = \begin{cases} 1 & \text{if } T(x) < c \\ \gamma & \quad\quad = \\ 0 & \quad\quad > \end{cases},$$

c & γ from the ~~above~~ size condn (does not depend on any particular $\theta_1 < \theta_0$)

Further ~~the~~ ϕ is non-increasing in T, $Q(\theta)$ is strictly decreasing in θ,

$\therefore$ the power fn $E_\theta \phi(T)$ is increasing in θ.

Next, we consider the following 2-sided hypotheses for the one-parameter exponential family distn with MLR in $T(x)$

$H_0: \theta \le \theta_1$ or $\theta \ge \theta_2$ ag. $H_A: \theta_1 < \theta < \theta_2$.

UMP(α) test is given by

$$\phi(T(x)) = \begin{cases} 1 & \text{if } c_1 < T(x) < c_2 \\ \gamma_j & T(x) = c_j,\ j=1,2 \\ 0 & T(x) < c_1 \text{ or } T(x) > c_2 \end{cases}$$

c_j, γ_j $j=1,2$ are $\ni$ $E_{\theta_1}(\phi(T)) = E_{\theta_2}(\phi(T)) = \alpha$.

Ex $x_1, \dots, x_n$ r.s. from $N(\mu, 1)$.

To test $H_0: \mu \le \mu_1$ or $\mu \ge \mu_2$ ag. $H_A: \mu_1 < \mu < \mu_2$.

The family of distn $N(\mu,1)$, $\mu \in \mathbb{R}$ has MLR in $\sum x_i$ (or, $\bar{x}$), say, $T(x)$

UMP(α) test is given by, $\phi(T(x)) = \begin{cases} 1 & \text{if } c_1 < \sum x_i < c_2 \\ 0 & \sum x_i \le c_1 \text{ or } \sum x_i \ge c_2 \end{cases}$

Here, $\gamma_1 = \gamma_2 = 0$, $c_1, c_2 \ni P_{\mu_1}(c_1 < \sum X_i < c_2) = P_{\mu_2}(c_1 < \sum X_i < c_2) = \alpha$, ie,

$$P_{\mu_1}\left[\frac{c_1 - n\mu_1}{\sqrt{n}} < \frac{\sum X_i - n\mu_1}{\sqrt{n}} < \frac{c_2 - n\mu_1}{\sqrt{n}}\right] = P_{\mu_2}\left[\frac{c_1 - n\mu_2}{\sqrt{n}} < \frac{\sum X_i - n\mu_2}{\sqrt{n}} < \frac{c_2 - n\mu_2}{\sqrt{n}}\right] = \alpha$$

$$\Rightarrow \left.\begin{aligned} \Phi\left(\frac{c_2 - n\mu_1}{\sqrt{n}}\right) - \Phi\left(\frac{c_1 - n\mu_1}{\sqrt{n}}\right) &= \alpha \\ \Phi\left(\frac{c_2 - n\mu_2}{\sqrt{n}}\right) - \Phi\left(\frac{c_1 - n\mu_2}{\sqrt{n}}\right) &= \alpha \end{aligned}\right\} \text{ 2 eqns in 2 unknowns } c_1 \,\&\, c_2$$

What about the 2 other types of 2-sided hypotheses, viz.,

$H_0: \theta = \theta_0$ ag. $H_A: \theta \ne \theta_0$

?? and $H_0: \theta_1 \le \theta \le \theta_2$ ag. $H_A: \theta < \theta_1$ or $\theta > \theta_2$. ?